Functions & Transformations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Evaluating functions**

If $f(x) = 3x - 4$, find $f(5)$.

$f(5) = 3(5) - 4 = 11$.

2 **[Foundational]** • **Target: Evaluating functions**

If $f(x) = x^2 - 2x + 1$, find $f(-2)$.

$f(-2) = 4 + 4 + 1 = 9$.

3 **[Foundational]** • **Target: Evaluating functions**

If $g(x) = \sqrt{x + 5}$, find $g(4)$.

$g(4) = \sqrt{9} = 3$.

4 **[Foundational]** • **Target: Evaluating functions**

If $h(x) = \frac{2x}{x-1}$, find $h(3)$.

$h(3) = \frac{6}{2} = 3$.

5 **[Foundational]** • **Target: Domain and range**

State the domain of $f(x) = \sqrt{x - 3}$.

$x - 3 \geq 0$, so the domain is $x \geq 3$.

6 **[Foundational]** • **Target: Domain and range**

State the domain of $f(x) = \frac{1}{x-5}$.

All real numbers except $x = 5$ (where the denominator is zero).

7 **[Foundational]** • **Target: Domain and range**

State the domain of $f(x) = \sqrt{7 - x}$.

$7 - x \geq 0$, so the domain is $x \leq 7$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Domain and range

State the range of $f(x) = x^2 + 2$.

Since $x^2 \geq 0$ for all real x, the range is $f(x) \geq 2$.

9 [Intermediate] • Target: Transformations: shifts

The graph of $y = x^2$ is shifted 3 units right and 2 units down. Write the new equation.

$y = (x - 3)^2 - 2$.

10 [Intermediate] • Target: Transformations: shifts

The graph of $y = |x|$ is shifted 4 units left and 1 unit up. Write the new equation.

$y = |x + 4| + 1$.

11 [Intermediate] • Target: Transformations: shifts

Describe the transformation that takes $y = x^2$ to $y = (x + 2)^2 - 5$.

A shift 2 units left and 5 units down.

12 [Intermediate] • Target: Transformations: reflections and stretches

The graph of $y = x^2$ is reflected over the x-axis and stretched vertically by a factor of 3. Write the new equation.

$y = -3x^2$.

13 [Intermediate] • Target: Transformations: reflections and stretches

Describe the transformation that takes $y = \sqrt{x}$ to $y = 2\sqrt{x}$.

A vertical stretch by a factor of 2.

14 [Intermediate] • Target: Transformations: reflections and stretches

Describe the transformation that takes $y = x^2$ to $y = -(x - 1)^2 + 4$.

A reflection over the x-axis, then a shift 1 unit right and 4 units up.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Composite functions • Cross-Lesson: 9.01 Statistics & Probability

Let $f(x) = 2x + 1$ and $g(x) = x^2 - 3$. Find $(f \circ g)(2)$.

$g(2) = 4 - 3 = 1$. $f(1) = 2(1) + 1 = 3$.

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16 [Challenging] • Target: Composite functions • Cross-Lesson: 1.01 Algebra & Equations

Using the same f and g, find $(g \circ f)(2)$.

$f(2) = 5$. $g(5) = 25 - 3 = 22$.

17 [Challenging] • Target: Composite functions • Cross-Lesson: 3.01 Sequences & Series

Using the same f and g, find a general formula for $(f \circ g)(x)$.

$f(g(x)) = 2(x^2 - 3) + 1 = 2x^2 - 6 + 1 = 2x^2 - 5$.

18 [Challenging] • Target: Composite functions • Cross-Lesson: 4.01 Trigonometry

Using the same f and g, find a general formula for $(g \circ f)(x)$.

$g(f(x)) = (2x + 1)^2 - 3 = 4x^2 + 4x + 1 - 3 = 4x^2 + 4x - 2$.

19 [Challenging] • Target: Inverse functions • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Find the inverse of $f(x) = 3x - 2$.

$y = 3x - 2 \Rightarrow x = 3y - 2 \Rightarrow y = \frac{x+2}{3}$. So $f^{-1}(x) = \frac{x+2}{3}$.

20 [Challenging] • Target: Inverse functions • Cross-Lesson: 6.01 Limits & Derivatives

Find the inverse of $f(x) = \frac{x-4}{5}$.

$y = \frac{x-4}{5} \Rightarrow 5y = x - 4 \Rightarrow x = 5y + 4$. So $f^{-1}(x) = 5x + 4$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Domain and range

The graph shows $f(x)$ defined only where plotted. State its domain.

a. $[-3, 3]$ ✓

b. All real numbers

c. $[-4, 4]$

d. $[0, 3]$

The curve is only plotted for x between -3 and 3.

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22 [Foundational] • Target: Even and odd functions

The graph shown is symmetric about the y-axis. Is the function even, odd, or neither?

a. Odd

b. Neither

c. Cannot be determined

d. **Even** ✓

Symmetry about the y-axis is exactly the definition of an even function.

23 [Foundational] • Target: Reading a function graph

The graph shows $y = f(x) = x^2 - 9$. Find $f(3)$ by reading the graph.

a. -9

b. 6

c. 9

d. 0 ✓

The curve crosses the x-axis at $x = 3$, so $f(3) = 0$.

24 [Foundational] • Target: Reading a function graph

Using the same graph, find the values of x where $f(x) = 0$ (the x-intercepts).

a. $x = 0$ and $x = 9$

b. $x = -9$ and $x = 9$

c. $x = 3$ only

d. $x = -3$ **and** $x = 3$ ✓

The curve crosses the x-axis at $x = -3$ and $x = 3$.

25 [Foundational] • Target: Reading a function graph

Using the same graph, find the y-intercept.

a. -3

b. 9

c. 0

d. -9 ✓

At $x = 0$, the curve crosses the y-axis at $y = -9$.

26 [Foundational] • Target: Evaluating functions

If $f(x) = 2x^2 + 1$, find $f(3) - f(1)$.

a. 18

b. 8

c. 16 ✓

d. 22

$f(3) = 19$, $f(1) = 3$; $19 - 3 = 16$.

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27 [Foundational] • Target: Transformations: shifts

The graph of $y = |x|$ is shifted 2 units left and 4 units up. Write the new equation.

a. $y = |x + 4| + 2$

b. $y = |x + 2| + 4$ ✓

c. $y = |x + 2| - 4$

d. $y = |x - 2| + 4$

Left 2: replace x with $x + 2$; up 4: add 4.

28 [Foundational] • Target: Transformations: shifts

Describe the transformation that takes $y = x^2$ to $y = (x - 3)^2 + 2$.

a. Shift right 2, up 3

b. Shift right 3, down 2

c. Shift left 3, up 2

d. **Shift right 3, up 2** ✓

Replacing x with $x - 3$ shifts right 3; adding 2 shifts up 2.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Transformations: shifts

The graph of $y = \sqrt{x}$ is shifted 6 units right. Write the new equation.

a. $y = \sqrt{x + 6}$

b. $y = \sqrt{x} - 6$

c. $y = \sqrt{x} + 6$

d. $y = \sqrt{x - 6}$ ✓

Right 6: replace x with $x - 6$.

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30 [Intermediate] • Target: Transformations: reflections and stretches

Describe the transformation that takes $y = x^2$ to $y = -(x+2)^2 + 1$.

a. Reflect over x-axis, shift right 2, down 1

b. Reflect over y-axis, shift right 2, up 1

c. **Reflect over x-axis, shift left 2, up 1** ✓

d. Shift left 2, up 1 (no reflection)

The negative sign reflects over the x-axis; $+2$ shifts left 2; $+1$ shifts up 1.

31 [Intermediate] • Target: Transformations: reflections and stretches

The graph of $y = \sqrt{x}$ is reflected over the y-axis. Write the new equation.

a. $y = -\sqrt{-x}$

b. $y = -\sqrt{x}$

c. $y = \sqrt{-x}$ ✓

d. $y = \sqrt{x}$

Reflecting over the y-axis replaces x with $-x$: $y = \sqrt{-x}$.

32 [Intermediate] • Target: Composite functions

Let $f(x) = x + 4$ and $g(x) = 2x$. Find $(f \circ g)(5)$.

a. 10

b. 9

c. 14 ✓

d. 18

$g(5) = 10$; $f(10) = 14$.

33 [Intermediate] • Target: Inverse functions

Find the inverse of $f(x) = x^3 - 2$.

a. $f^{-1}(x) = \sqrt[3]{x} + 2$

b. $f^{-1}(x) = (x+2)^3$

c. $f^{-1}(x) = \sqrt[3]{x-2}$

d. $f^{-1}(x) = \sqrt[3]{x+2}$ ✓

$y = x^3 - 2 \Rightarrow x = \sqrt[3]{y+2}$.

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34 [Intermediate] • Target: Inverse functions

Find the inverse of $f(x) = 5x + 10$.

a. $f^{-1}(x) = \frac{x-10}{5}$ ✓

b. $f^{-1}(x) = \frac{x+10}{5}$

c. $f^{-1}(x) = \frac{x}{5} - 10$

d. $f^{-1}(x) = 5x - 10$

$y = 5x + 10 \Rightarrow x = \frac{y-10}{5}$.

35 [Intermediate] • Target: Piecewise functions

A function is defined by $f(x) = 2x$ for $x \leq 0$, and $f(x) = x^2 + 1$ for $x > 0$. Find $f(0) + f(3)$.

a. 1

b. 11

c. 0

d. 10 ✓

$f(0) = 0$; $f(3) = 9 + 1 = 10$; sum $= 10$.

36 [Intermediate] • Target: Even and odd functions

Determine whether $f(x) = x^3 - 4x$ is even, odd, or neither.

a. Neither

b. Both even and odd

c. Even

d. **Odd** ✓

$f(-x) = -x^3 + 4x = -f(x)$: odd.

37 [Intermediate] • Target: Even and odd functions

Determine whether $f(x) = x^2 + 3x$ is even, odd, or neither.

a. Both even and odd

b. **Neither** ✓

c. Odd

d. Even

$f(-x) = x^2 - 3x$, which is neither $f(x)$ nor $-f(x)$: neither.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Operations on functions • Cross-Lesson: 4.01 Trigonometry

Let $f(x) = 2x$ and $g(x) = x + 6$. Find $(f - g)(x)$.

a. $3x + 6$

b. $x - 6$ ✓

c. $x + 6$

d. $-x - 6$

$2x - (x + 6) = x - 6$.

39 [Challenging] • Target: Word problems: functions in context • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

A company's daily profit (in dollars) from selling x units is modeled by $P(x) = -2x^2 + 80x - 300$. Find the profit when 15 units are sold.

a. 450 **dollars** ✓

b. 150 dollars

c. 600 dollars

d. 900 dollars

$P(15) = -2(225) + 1{,}200 - 300 = 450$ dollars.

40 [Challenging] • Target: Word problems: functions in context • Cross-Lesson: 6.01 Limits & Derivatives

The Fahrenheit conversion of a Celsius temperature is $F(C) = \frac{9C}{5} + 32$. Find $F(20)$.

a. 72°F

b. 68°**F** ✓

c. 36°F

d. 52°F

$F(20) = \frac{9(20)}{5} + 32 = 36 + 32 = 68$.

41 [Challenging] • Target: Word problems: functions in context • Cross-Lesson: 7.01 Integration & Applications

A ball's height (in meters) t seconds after being thrown is $h(t) = -5t^2 + 20t + 2$. Find the height at $t = 2$ seconds.

a. 2

b. 18

c. 22 ✓

d. 42

$h(2) = -5(4) + 40 + 2 = 22$ meters.

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42 [Challenging] • Target: Functions & Transformations • Cross-Lesson: 8.01 Exponentials & Logarithms

If $f(x) = x^2 - 6x - 7$, find $f(6)$.

a. -7 ✓

b. 7

c. -10

d. -4

$f(6) = (6)^2 - 36 - 7 = -7$.

43 [Challenging] • Target: Functions & Transformations • Cross-Lesson: 9.01 Statistics & Probability

If $f(x) = -3x^2 - 5x + 9$, find $f(-3)$.

a. 0

b. -6

c. -3 ✓

d. 3

$f(-3) = -3(-3)^2 + 15 + 9 = -3$.

44 [Challenging] • Target: Functions & Transformations • Cross-Lesson: 1.01 Algebra & Equations

If $f(x) = -x^2 + 7x + 1$, find $f(-1)$.

a. -4

b. 7

c. -7 ✓

d. -10

$f(-1) = -(-1)^2 - 7 + 1 = -7$.

45 [Challenging] • Target: Functions & Transformations • Cross-Lesson: 3.01 Sequences & Series

If $f(x) = -3x^2 - x - 1$, find $f(4)$.

a. -56

b. 53

c. -50

d. -53 ✓

$f(4) = -3(4)^2 - 4 - 1 = -53$.

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